

TEJESWARA REDDY DUDYALA

9110727080 • dtejaswarreddy10@gmail.com • Kadapa, India 516175

[LinkedIn: linkedin.com/in/tejeswara-reddy-dudyala-74784821b](https://www.linkedin.com/in/tejeswara-reddy-dudyala-74784821b)

Experience - 2 Years 7 Months

Design Engineer

Nov 2025 - Present

ACL Digital [Client - AMD]

Hyderabad

- Creation of Test Plan and Written RTL to verify the New Features. Developed AI Agent to Review Test Plan.
- Responsible for Two Regression Suites and Working on the Automation of Regression Suit Maintenance using AI.
- Built a regression automation framework with AI to map new test suites to actual regression runs and generate execution logs/reports, and email notifications for faster validation tracking

RTL Design Engineer

Jan 2024 - July 2025

Krisemi Design Technologies[1 Year 7 Months]

Bengaluru

- Understanding of each module and RTL Upgrade of 3 modules from DVB-S2 to DVB-S2X version.
- Written FSM-based RTL for the modules present in the DVB-S2 protocol.
- Verified each module of DVB-S2 & DVB-S2X in Vivado Simulation.

Design Verification Trainee

June 2023 - November 2023

Moschip Institute of Silicon Systems[6 Months]

Hyderabad

- Designing of 16 bit ALU and 8 bit Customized up-down counter using Verilog and Verification using SV.
- Verification of APB based 10/100 ETHERNET and AMBA-3 APB Protocol by building the UVM Verification architecture from scratch to the functional coverage and ABV.

Skills

HDL	: Verilog, VHDL
HDVL	: System Verilog
EDA Tools	: Questasim, VCS, Vivado, Onespin, Riviera, Xcelium
Protocols Worked on	: APB3, 10/100 APB Based ETHERNET, DVB_S2 and DVB_S2X, UART
Protocols Known	: AXI4, SPI, DMA, FIFO
Scripting	: Python, TCL with the help of LLM's
Concepts	: FSM Based Designing , Lint , CDC, Synthesis, Clock Gating, ASIC FLOW
Version Control	: Github, Perforce
AI & Automation Tools	: Claude Code (CLI), Claude Opus, OpenAI Codex, DEVStudio, Agentic Flows.

Projects

In-Memory Computing (IMC) SRAM Engine [1 Month]

March 2025

- Designed a 256×32 dual-port SRAM-based IMC engine in Verilog to accelerate edge AI/ML inference by performing bitwise computations directly within memory, addressing the von Neumann memory-wall bottleneck.
- Verified the design with self-checking testbenches (directed + randomized tests) and synthesized for resource utilization and throughput analysis.

MIPS / RISC-V Processor Design (Single-Cycle & Pipelined)- [3 Months]

Sept 2025 - Nov 2025

- Designed a 32-bit MIPS/RISC-V processor supporting R, I and J-type instruction formats.
- Implemented both single-cycle and 5-stage pipelined architectures (IF, ID, EX, MEM, WB).
- Added hazard detection, data forwarding logic to ensure correct pipelined execution.

- Developed ALU, Register File, Control Unit, PC, Branch logic, Memory Interface modules.
 - Verified processor execution using waveform analysis in ModelSim/QuestaSim.
- UART TX/RX Design (With & Without FIFO Support) - [1 Month]** July 2025 - Aug 2025
- Designed Tx and Rx modules with configurable baud rate, data bits, parity, and stop bits.
 - Implemented two architectures: direct TX/RX datapath and FIFO-based UART for improved throughput and reduced CPU load.
 - Developed sync FIFO using circular buffer logic to enable buffered serial communication.
 - Performed loopback testing to validate end-to-end TX→RX data integrity using QuestaSim.
- Development Of DVB - S2 and S2X IP -Inhouse Project - [1Y 7M]** Jan 2024 - July 2025
- Designed Mode Adaptation Module in the DVB-S2 IP (Opted FSM based design).
 - Designed CRC8 calculator for 8 bit input.
 - Understanding the DVB-S2 and DVB_S2X Protocols.
 - Upgraded the 3 modules(Frame_length_enforcer,LDPC,PL) RTL from DVB-S2 version to DVB-S2X version
 - Module by Module Verification of DVB-S2 & DVB-S2X by Simulation on Vivado.
 - Total DVB-S2 and DVB-S2X System Verification
- APB based ETHERNET Protocol - Internal Project - [3 Months]** Oct 2023 - Dec 2023
- Understanding the Ethernet architecture and functionality.
 - Preparation of the Verification Plan, Test plan, Coverage plan and Assertion plan.
 - Involving the IP level verification using UVM environment for Ethernet IP Core.
 - Implementation of the test scenarios to verify the functionality of Ethernet IP Core.
 - Code coverage and Functional coverage analysis and assertion based formal verification.
- AMBA 3 APB Protocol - Internal Project - [1 Month]** Sept 2023 - Oct 2023
- Understanding the APB protocol specification.
 - Preparation of the Verification Plan, Test plan, Coverage plan and Assertion plan.
 - Protocol verification with a single master and single slave in both SV and UVM.
 - Implementation of the Functional coverage model and Assertions.
- UP-DOWN COUNTER - Internal Project - [1 Month]** Aug 2023 - Sept 2023
- Understanding the specification.Implemented the up down Design.
 - Verified the Design in SV Environment.Prepared the test, verification and coverage plan.
 - Implemented Coverage and Assertions to check the functionality of DUT.
- ALU - Internal Project - [1 Month]** July 2023 - Aug 2023
- Understanding the specification..Implemented the ALU design.
 - Verified with a basic test bench.Prepared the design plan.
- Control and Analysis of BLDC Motor using Different Control Techniques**
- In this project the speed control performance of BLDC motor is analysed using different control techniques in MATLAB software.
- Switching Control of Home Appliances by IR Remote using Microcontroller**
- We designed a low-cost home automation circuit using micro controllers and relays.

Education

Birla Institute of Science and Technology,Pilani	Online
Master of Technology - Work Integrated Learning Program	Jan 2025 - Dec 2026
Coursework: VLSI Design & Microelectronics	Ongoing
R V R & J C College of Engineering	Guntur,AP
Bachelor of Technology - Full Time	Aug 2019 - April 2023
Coursework: Electrical and Electronics Engineering	CGPA : 9.13